

SECTION – A

Answer any FIVE of the following Questions.

5 X 5 = 25M

1. What is Unsupervised Learning.
2. Define Clustering.
3. What is Associative analysis.
4. How to mine a click stream from a news site.
5. Explain mining frequent from an FP-tree
6. What are Dimensionality reduction techniques.
7. List Applications of the SVD.
8. Explain Matrix factorization.

SECTION – B

Answer ALL the Following Questions.

5 X 10 = 50M

9. a) Discuss k-means clustering algorithm.
(OR)
b) How to improve cluster performance with post processing.
10. a) Explain Apriori algorithm along with its principles.
(OR)
b) Discuss Mining association rules from frequent item sets.
11. a) Define Finding frequent item sets: FP-growth –FP trees, Build FP-tree
(OR)
b) List out steps to find co-occurring words in a Twitter feed
12. a) Discuss Principal component analysis to reduce dimensionality.
(OR)
b) How PCA is used to reduce the dimensionality of semiconductor manufacturing data
13. a) Discuss how Singular value decomposition(SVD) is implemented in Python.
(OR)
b) Discuss Collaborative filtering–based recommendation engines.